Importance of Cryptography: HTTP vs HTTPS Credential Transmission

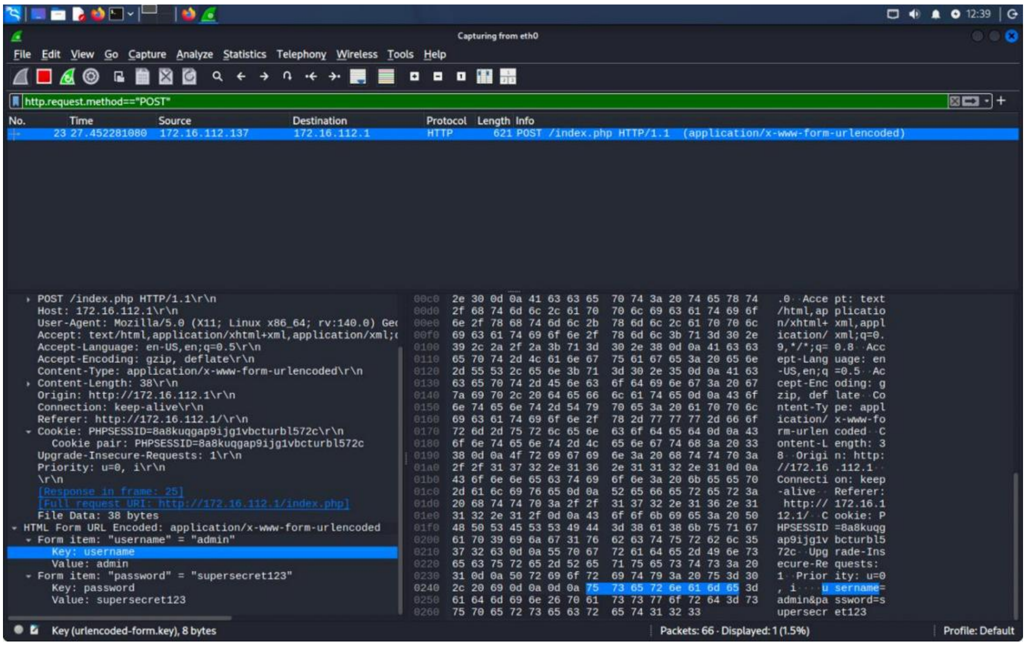
Objective

For this practical, I could visualise what happens to login credentials when they are sent over HTTP versus HTTPS, rather than just reading about it. I used Wireshark to capture live traffic to compare an unencrypted connection with an encrypted one side by side.

Procedure

I entered dummy login details into a test login page over HTTP, and then did the same thing on an HTTPS-secured page. While doing this, I had Wireshark running in the background, capturing traffic on the eth0 interface. Once I had both captures, I went through the packets to check whether the username and password could actually be read.

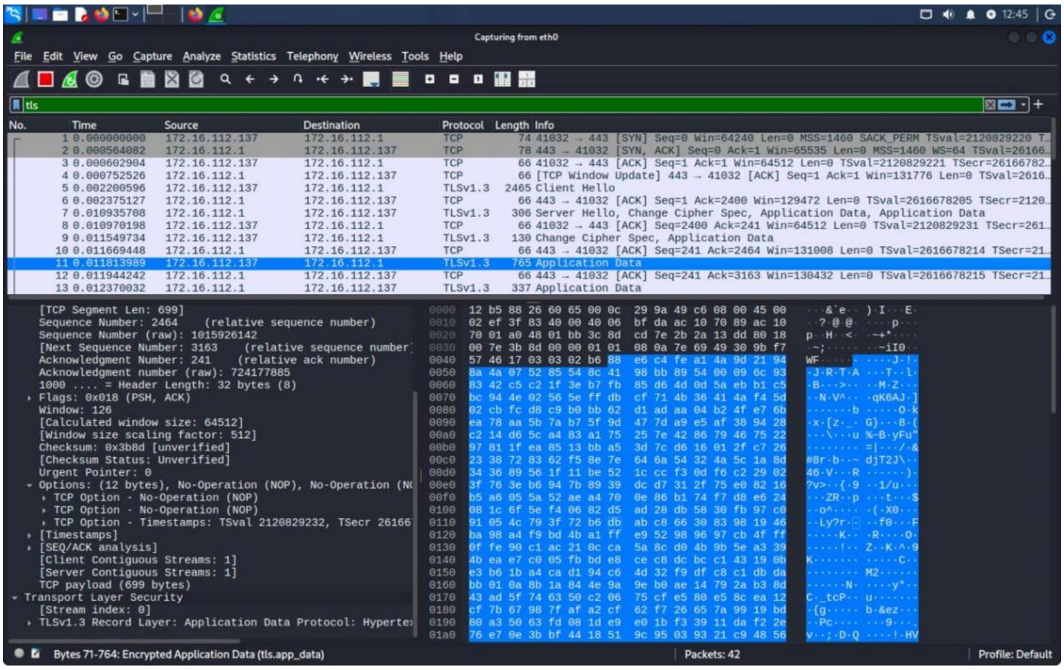
1. HTTP Traffic Analysis



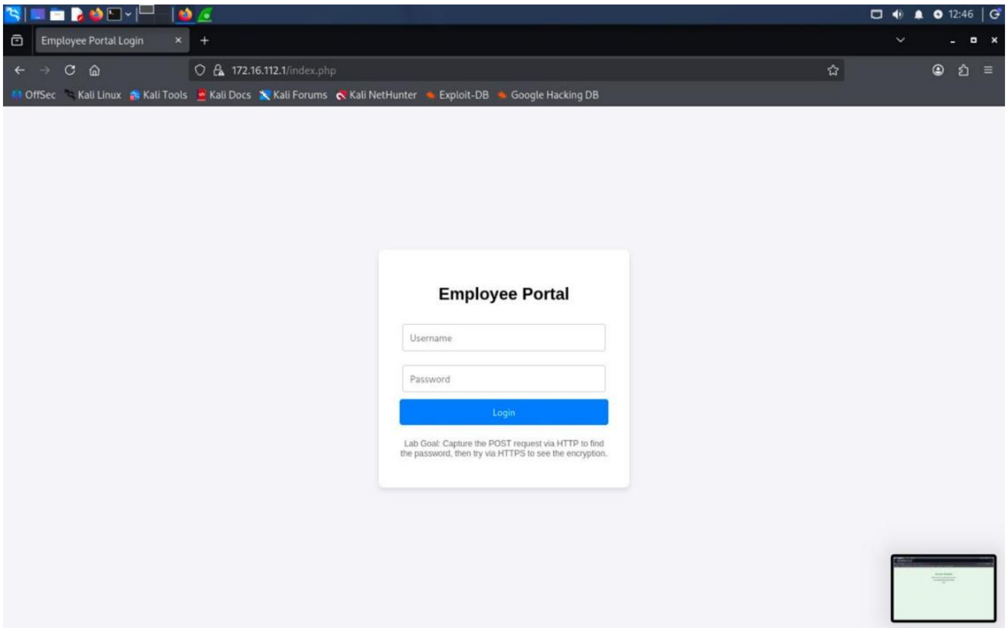
I captured packet traffic on the eth0 interface while logging into two different sites to compare how they handle credentials over the network.

I started a Wireshark capture, then opened a test login page that uses plain HTTP. I typed in some dummy credentials, hit submit, and stopped the capture. Looking through the packets, the POST request stood out immediately that the username and password were sitting right there in the HTML form data, completely readable in plain text.

2. HTTPS Traffic analysis



I repeated the same process, but this time against GitHub's login page, which runs over HTTPS. Same steps: start capture, login with dummy credentials, submit, stop capture. This time, though, the login data was nowhere to be found in readable form. TLS encryption wraps the traffic before it ever leaves the machine, so anyone watching the wire just sees ciphertext instead of a username and password.



3. Comparison

Entity	**HTTP**	**HTTPS**
URL indicator	Starts with http://	Starts with https://
Browser security indicator	No padlock / "Not Secure" warning	Padlock icon shown in address bar
SEO / ranking impact	Can be penalized by search engines	Preferred by search engines (e.g., Google)
Certificate requirement	Not required	Requires SSL/TLS certificate from a CA
Performance overhead	Slightly faster (no encryption overhead)	Slightly slower due to encryption/decryption (negligible with modern hardware)

4. Role of Encryption

Doing this practical really helped me understand why encryption matters so much when sensitive data is involved. With HTTP, credentials go out as plain text, so anyone watching the network could read them without any extra effort. HTTPS fixes this by encrypting everything with TLS before it is sent, so even captured packets are unreadable. On top of that, TLS also gives you integrity, meaning you can tell if data has been tampered with in transit, and authentication, which confirms that are actually talking to the real server and not an imposter.

5. Conclusion

This experiment made the difference between HTTP and HTTPS very clear to me. Seeing the dummy credentials sitting in plain text during the HTTP capture was a good reminder of just how exposed data can be without encryption. The HTTPS capture, on the other hand, kept the same kind of sensitive information completely protected. I could not read anything meaningful from the packets. Overall, this showed me why cryptography is so essential for protecting data as it travels across a network, and why HTTPS should always be used instead of HTTP whenever login credentials, personal information, payment details, or any other sensitive data are being sent.